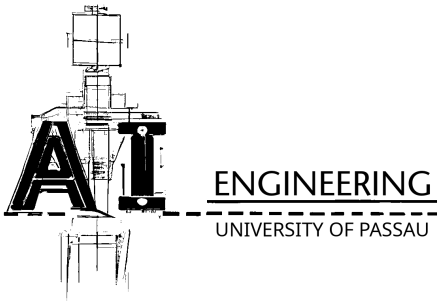
Mining Software Repositories

Exercise 3

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General information (read carefully!)

With the first set of exercises, we have started to build a set of tools for data mining. During the upcoming exercises, we will transition towards more investigative tasks. Those might require you to quickly adjust your current data mining scripts in order to focus on answering research questions.

Task 1: Storing data in a harmonized database

1.1 Designing a harmonized database for exercise 1 and 2

In the past two exercises we have been dealing with the extraction of data from three sources: Git, Jira and GitHub. But we have not yet harmonized those disparate sources. For this, we need to make sure that we deal with duplicates while also mapping the data to a common structure.

Revisit the previous exercises and remember your observation during the exploration phase.

- Devise a plan to store commits (including changed files, old and new path, and the type of change), issues (from both Jira and GitHub), authors (from all sources) and pull requests inside a harmonized database.
- Choose a database management system (DBMS). Be prepared to reason about your choice.
- Visualize your plan with an entity relationship diagram.

1.2 Creating a harmonized database

It is now time to put your plan into action. You should reuse data you have mined in the previous exercises or adjust your existing scripts to obtain the required data to speed up the process.

1. Create an instance of the DBMS that you chose in the previous section. Make sure that you also install an administration interface so that you can inspect the database later.
2. Store the data of the Apache Ignite[1] project inside your harmonized database. Either by adjusting the scripts of the previous exercises and writing directly into the database,

[1] Ignite: `https://github.com/apache/ignite`

or by loading the data from those scripts and transferring to the database from there. Make sure that all the fields of your design are filled with data.

Recommended Tools

2.1 Diagrams.net

Entity relationship diagrams are a common way for visualizing database structures.[2] You can use a tool like Diagrams.net to create the diagram.[3]

2.2 Database management systems

There are various database management systems. For this exercise, we recommend using MariaDB.[4]

2.2.1 Without docker

If you have no experience with docker, install MariaDB directly in your operating system. Complete the installation process from the download page[5] on Windows, or install MariaDB through the command line on Mac (Homebrew[6]) or Linux. If prompted, set a password that you can remember.

2.2.2 With docker

If you have experience with docker, we would recommend to set up an instance with the official docker-compose file from docker hub.[7] With that, you also already have an instance of adminer, which offers a simple user interface for querying the database.

2.3 Database administration software

One option for a GUI database administration software is DBeaver. It runs on Windows, Mac and Linux and can be used for free.[8]

[2] Entity Relationship Diagrams: `https://en.wikipedia.org/wiki/Entity-relationship_model`
[3] Diagrams.net: `https://app.diagrams.net`
[4] MariaDB: `https://mariadb.com`
[5] MariaDB Download: `https://mariadb.com/downloads/`
[6] MariaDB Homebrew: `https://mariadb.com/kb/en/installing-mariadb-on-macos-using-homebrew/`
[7] MariaDB Docker Hub: `https://hub.docker.com/_/mariadb`
[8] DBeaver: `https://dbeaver.io`